

Luke Wang

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EDUCATION

University of California, Berkeley | Berkeley, CA

Bachelor of Art in Computer Science | August 2023 – May 2027 | GPA: 3.99/4.00

Intended to pursue an M.S in Computer Science after graduation

Course: Machine Learning, Deep Learning, Language Modeling, Efficient Algorithms, Probability, Optimization Models, Internet Architecture, Data Engineering

SKILLS

Programming Languages: Python, C++, Java, C#, SQL, HTML, TypeScript, JavaScript

Machine Learning: PyTorch, scikit-learn, Transformers, Diffusion Models, Flow Matching, Random Forest, Fine-Tuning

Systems & Data: Docker, Kubernetes, Linux, Git, LangGraph, NumPy, pandas, PostgreSQL, Redis, Agent SDK

EXPERIENCE

Berkeley AI Research Speech Group | Berkeley, CA

Research Assistant | May 2026 – Present | Advised by Gopala Anumanchipalli

- Built an **audio-to-audio speech codec** in PyTorch speech using interpretable, human-inspired articulatory representations and a neural vocoder competing with larger neural-codec in internal benchmarks
- Engineered experiment orchestration across **8 GPU** with atomic checkpoints, bit-exact resume, and automated evaluation
- Scaled LibriTTS training pipelines to **522 hours and 334K** utterances using memory-mapped storage and length-aware batching, with automated alignment and validation to catch supervision errors
- Developed robotics motion control model using **Dynamic Movement Primitives (DMPs)** and spring-damper dynamics, generating articulatory trajectories with explicit control over movement targets and stiffness
- Developed **prompt conditioned flow matching for text-to-speech** for audio variations in **accent and impression**

Accenture | San Francisco, CA

Software LLM Development Summer Analyst | May 2026 - Present

- Built an **Agentic RPA migration** with **LangGraph** to convert Blue Prism to UiPath, achieving a 99.8% verifiable success rate on 1000-case benchmark
- Engineered **SDK workflows** executed independent migrations concurrently, reduced repeated rebuilds and token cost in code generation and validation.
- Implemented restart-safe execution using serialized LangGraph state and **PostgreSQL checkpoints**, preserving migration progress across process failures and **human-approvals**.
- Built **sandboxed environment** to validate migrated workflows, detecting business-logic errors through edge-case testing

Calpine | Houston, TX

Software Engineering Intern | May 2025 – August 2025

- Built automated **ML data pipelines** with C#, .NET, and SSIS, integrating SQL Server and Azure ADLS Gen2 to supply datasets for **energy-forecasting models**
- Replaced manual data-transfer steps with automated ETL jobs, cutting **transfer time by 90%** for downstream forecasts
- Built archival and restore pipeline to **ADLS**, reducing historical-data storage by 95% while preserving recoverability
- Parameterized ETL by time range, etc. and introduced **database snapshots** to reduce load on the primary SQL Server

PROJECTS

PillarFortune — Conversational AI Application | React, TypeScript, Cloudflare Workers

- Built a full-stack conversational tarot application with personalized readings and follow-ups, integrating a React frontend
- Managed session with Durable Objects and persisted contexts, follow-ups on original question, drawn cards, and dialogue
- Separated deterministic card selection from LLM interpretation, adding JSON schema and structured card reference validation with bounded retries and fallback handling for invalid outputs.

Distributed Task Scheduling System | Java, Spring Boot, Redis, RabbitMQ, PostgreSQL, Docker, Kubernetes

- Built distributed task scheduling system using **Java** and **Spring Boot** handling **20K+ daily tasks** reliably
- Implemented message queue with **RabbitMQ** supporting priority scheduling and retry mechanisms for task failures
- Designed task metadata storage with **PostgreSQL** and **Redis** caching reducing average query latency by **50%**
- Deployed with **Docker** and **Kubernetes** achieving horizontal scaling and **99.99% uptime** with automated failover

LLM Fine-Tuning & Evaluation | PyTorch, Hugging Face, PEFT

- Built a supervised fine-tuning and evaluation pipeline for technical multiple-choice questions with Qwen2.5
- Benchmarked LoRa against the unmodified model on accuracy; profiled training throughput and peak GPU memory
- Evaluated shuffled answers to measure option order sensitivity and distinguish invalid outputs from incorrect answers